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Recurring Natural Disasters and Macroeconomic Scarring:

A Risk-Adjusted DIGNAD Framework

Giovanni Melina, Mehdi Raissi, Azar Sultanov, and Bruno Versailles

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Prepared by Giovanni Melina, Mehdi Raissi, Azar Sultanov, and Bruno Versailles

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ABSTRACT: Recurring climate-related natural disasters pose significant macroeconomic and fiscal risks for highly exposed countries. We extend the Debt, Investment, Growth, and Natural Disasters (DIGNAD) model to incorporate Annual Average Loss (AAL) metrics, capturing the cumulative macro-fiscal effects of repeated natural disaster shocks. The augmented DIGNAD framework introduces recurring annual capital losses and negative productivity shocks, yielding more realistic growth and public debt trajectories under persistent disaster risk. We quantify the GDP losses from recurrent natural disasters and examine the trade-offs between higher adaptation investment (leading to higher recovered growth) and public debt accumulation. Using cross-country simulations for 33 disaster-prone economies and case studies (the Philippines, Mauritania, and Barbados), we find substantial heterogeneity in disaster impacts and adaptation effectiveness. Comparing risk-adjusted baseline projections with scenarios of scaled-up resilient vs. standard infrastructure investment, we find that investing in resilient public capital can significantly mitigate output losses in the long-term, albeit at the cost of higher public debt.

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WORKING PAPERS

Recurring Natural Disasters and Macroeconomic Scarring:

A Risk-Adjusted DIGNAD Framework

Prepared by Giovanni Melina, Mehdi Raissi, Azar Sultanov, and Bruno Versailles[1]

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1. Introduction

Climate-related natural disasters are increasing in frequency and severity (IPCC, 2023), posing significant macroeconomic and fiscal risks for many low-income countries (LICs) and emerging markets (EMs). These shocks—ranging from tropical cyclones and floods to droughts—can destroy physical capital, suppress economic growth, widen fiscal deficits, and raise public debt. Yet, current macroeconomic projections typically consider only extreme one-off disasters via stress tests, focusing on rare high-impact events. While such stress tests are useful for gauging tail risks, they overlook the cumulative impacts of recurrent natural disasters in highly exposed economies.

Standard medium-term projections often assume an unchanged climate baseline and exclude probabilistic disaster risk modeling. As a result, the expected impacts of events with probabilities ranging from 10% (once in 10 years) to 0.01% (once in 10,000 years) are rarely reflected in growth and debt forecasts. This gap is particularly problematic for the 30+ disaster-prone LICs and EMs that experience frequent and/or severe natural disasters. Relying *only* on extreme-event stress tests means that the annual cumulative toll of repeated disasters remains hidden in macro-fiscal projections. This omission can impart an optimism bias to growth and debt projections, which may affect macro-fiscal assessments. Moreover, traditional macro-fiscal frameworks generally treat climate adaptation investments purely as a cost, without accounting for the growth and fiscal risk mitigation benefits such investments may yield (Mitra et al., 2025). These gaps point to the need for enhanced modeling approaches that better integrate recurring disaster risks and adaptive investments into macro-fiscal analysis.

The Debt, Investment, Growth, and Natural Disasters (DIGNAD) model has emerged as a workhorse at the IMF for analyzing macro-fiscal risks from one-off climate-related disasters, owing to its nonlinear dynamics and intricate fiscal-debt linkages (Marto et al., 2018).[2] This paper augments the DIGNAD model to incorporate disaster risk on a recurring, probabilistic basis rather than only one-off extreme shocks.[3] We utilize Annual Average Loss (AAL), a metric representing the expected annual loss from natural disasters (derived from the full probability distribution of hazard events) and integrate it into DIGNAD as a standing annual shock to TFP and capital. By embedding AAL-based disaster shocks each period, our model produces risk-adjusted projections that reflect the cumulative macro-fiscal impacts of recurring natural disasters on vulnerable economies.[4] In this way, we explicitly recognize that for many countries,

[2] DIGNAD is a dynamic general equilibrium model designed to assess the macroeconomic and debt impact of natural disasters and reconstruction policies.

[3] Cantelmo, Melina, and Papageorgiou (2023) develop a stochastic DSGE framework in which disaster risk affects macroeconomic outcomes and the stochastic steady state. In their framework, expectations about disaster risk affects decisions and the stochastic steady state even when no disaster is realized. Our objective is different and narrower: we retain DIGNAD's operational fiscal-debt architecture and use country-specific AALs to produce risk-adjusted projections and investment-policy comparisons across a broad set of disaster-prone economies. Relatedly, Melina and Santoro (2021) and IMF (2022) assess the effects of a sequence of disaster shocks.

[4] Throughout the paper, we use "recurring natural disasters" for the physical phenomenon, "AAL-based expected-loss shock" for the model input, and "risk-adjusted expected-loss path" for the model scenario.

disaster risk is not a hypothetical scenario but a chronic drag on long-run growth and public finances.[5] This drag reflects realized recurring disasters, unlike models where agents adjust behavior in anticipation of disasters even if these do not materialize (e.g., Fernández Corugedo et al, 2023).

Our approach bridges two strands of the literature: on the one hand there is the strand that focuses on extreme one-time disaster shocks in macroeconomic models (e.g., Felbermayr et al., 2022; Nguyen et al., 2026) and on the other hand there is the research on the cumulative effects of slow-moving temperature increases on long-run growth (e.g., Kahn et al., 2021). By modeling frequent natural disasters of different probabilities and intensities, we provide a comprehensive assessment of climate-related macro-fiscal vulnerabilities that lie between these extremes. We make three distinct contributions. First, we show how the AAL estimates can be mapped into DIGNAD to construct risk-adjusted expected-loss projections for a cross-country sample of 33 disaster-prone economies, thereby complementing traditional one-off disaster stress tests.[6] Second, we show that recurring disaster risk changes the behavior of macro-fiscal variables in ways that are not fully captured by one-time shock analysis: lower output levels progressively erode the revenue base, repeated reconstruction needs create persistent financing pressures, and debt ratios can worsen even without a single catastrophic event. Third, we compare, for the same public-investment envelope, standard infrastructure investment with resilient infrastructure investment, thereby isolating the macro-fiscal value of adaptation in an environment where disaster losses recur over time.

Our results reveal substantial heterogeneity across countries experiencing recurring natural disasters. Among the 33 disaster-prone economies in our sample, AAL estimates range from below 0.2% to nearly 10% of GDP per year, implying widely varying baseline risks. Median annual adaptation investment across the sample is 0.4 percent of GDP, but the variation is wide, ranging from 0.1 percent to more than 3 percent of GDP across countries. In a scenario with additional adaptation investments (financed by public borrowing), median cumulative GDP losses under recurring shocks are 25% lower than in a risk-adjusted baseline without adaptation over time. However, this comes at the cost of a higher public debt path—5 percentage points of GDP above the baseline by the simulation horizon. The output–debt trade-offs and the effectiveness of adaptation vary markedly across countries, influenced by factors such as initial public capital stocks, fiscal space, public investment efficiency, and borrowing costs. For instance, an emerging market like the Philippines exhibits a growth rebound and recovers a large share of disaster-induced output losses under a high adaptation effort, whereas some low-income countries (e.g., Lao PDR, Sudan) and small states (e.g., Belize, The Bahamas) realize only modest gains (offsetting only about 5–10% of losses) due to the interaction of high exposure, initial public capital and infrastructure investment assumptions, public investment efficiency, fiscal space, borrowing costs, and financing mix . These findings highlight the importance of well-targeted, high-yield adaptation investments (Hallegatte et al., 2019) and underscore that resilience strategies must be tailored to country-specific conditions.

[5] Traditional DIGNAD applications are well suited to one-off disaster stress tests, where the central question is how an economy absorbs, finances, and recovers from a realized tail event. By contrast, the AAL-based extension developed here provides a risk-adjusted expected-loss scenario that embeds the recurrent annual drag from disaster risk into macro-fiscal projections. Tail-event stress tests remain necessary, but they should be evaluated relative to projections that already incorporate expected recurring losses.

[6] The AAL-based path is a deterministic expected-loss scenario, not a stochastic simulation or substitute for tail-risk stress tests. It captures mean annual losses but not the timing or nonlinear effects of rare catastrophic events. The framework can be used to complement WEO-style baselines with risk-adjusted projection paths, inform support adaptation-investment analysis, and provide inputs into debt sustainability discussions.

Beyond the cross-country analysis, we provide three in-depth country case studies to illustrate the role of initial macroeconomic conditions and investment choices: (i) the Philippines (an emerging market with high exposure to tropical cyclones and relatively strong public financial management capacity), (ii) Mauritania (a low-income, arid Sahelian economy with recurrent droughts and floods), and (iii) Barbados (a small open island state highly exposed to natural disasters). These case studies demonstrate how our risk-adjusted projections can deepen the understanding of climate vulnerability beyond traditional one-off disaster analysis. We also relate our model outcomes to relevant literature, notably the detailed work on returns to adaptation from the World Bank's Country and Climate Development Reports (CCDRs) and recent country-specific studies to provide empirical context on high-frequency climate shocks, adaptation investments, and macro-fiscal policy trade-offs.

The remainder of the paper is organized as follows. Section 2 provides an overview of the original DIGNAD model and details the modifications needed to incorporate recurring disaster risk via AAL. Section 3 describes the data and calibration, including the derivation of country-specific AAL estimates and adaptation parameters. Section 4 presents the results: first an overview of the cross-country baseline and investment scenarios, followed by the three country case studies and associated trade-offs. The final section concludes with key takeaways and directions for future research.

2. Incorporating Recurring Natural Disasters into DIGNAD

Implementing the framework set out in the introduction requires several enhancements to the DIGNAD model of Marto, Papageorgiou, and Klyuev (2018). Their model, in turn, extends the DIG model of Buffie et al. (2012) to analyze the effects of Cyclone Pam on Vanuatu and to explore how the country could have enhanced its resilience. We outline below the key features of that 'original' DIGNAD model which we inherit and then explain what is new in our paper from a modeling perspective.

The DIG model is a real, dynamic, two-sector small open economy with non-tradable and tradable goods. Perfectly competitive firms in each sector use public capital alongside private capital and labor in production. The government can borrow externally and domestically, with fiscal instruments ensuring long-run debt sustainability. The extension incorporating natural disasters (DIGNAD) allows the government to invest in both standard infrastructure and adaptation (resilient) capital, as well as to accumulate assets in a savings fund. Adaptation capital is more resilient to climate-related natural disasters. It helps preserve standard infrastructure and mitigates the damage and losses inflicted on the economy. Public adaptation capital enters firms' production functions together with standard infrastructure, thereby also protecting private capital indirectly. Conversely, a savings fund (if available) immediately provides liquidity to start reconstruction in the aftermath of a disaster. If the government lacks such a fund, reconstruction costs must be financed by raising debt or taxes. Natural disasters in this setup damage public and private capital and temporarily reduce Total Factor Productivity (TFP). We next present the core equilibrium relationships of the baseline DIGNAD model, before turning to our modifications for recurring natural disasters of AAL size.

2.1 Baseline DIGNAD Structure

Firms: Time is discrete, indexed by *t*. There are two production sectors: non-tradables (n) and tradables (x). Production in each sector is Cobb–Douglas in private inputs and effective public capital. Formally, for each sector $j \in \{n, x\}$:

$$y_{j,t} = A_{j,t}\, \tilde{z}_{t-1}^{\psi_j} k_{j,t-1}^{\alpha_j} l_{j,t}^{1-\alpha_j},$$

where $y_{j,t}$ denotes gross output, $A_{j,t}$ is sectoral TFP, $k_{j,t}$ and $l_{j,t}$ are the private capital and labor employed in sector j, α_j is the private-capital share in production, $\tilde{z}_t$ is effective public capital input, and ψ_j is the output elasticity of sector j with respect to public capital.

A natural disaster shock reduces effective output. Let $D_{j,t} \in [0,1]$ be the disaster intensity in sector j (the fraction of that sector's output that would be destroyed by the shock absent mitigation). Public adaptation capital helps mitigate these losses: a higher stock of z_a (resilient capital) lowers the effective intensity of the shock. In the Marto et al. model, this is captured via parameters $\pi_j \geq 0$ and $\nu_D \geq 0$ scaling the protective effect of adaptation. Formally, the mitigated disaster intensity is given by

$$\widetilde{D}_{j,t} = \frac{D_{j,t}}{(1 + \pi_j z_{a,t-1})^{\nu_D}}$$

so that larger z_a (and/or bigger π_j and ν_D) implies a smaller fraction of output lost for a given raw intensity $D_{j,t}$.

Public capital and adaptation: Effective public capital $\tilde{z}_t$ is modeled as a CES aggregate of standard infrastructure capital $z_{i,t}$ and adaptation (resilient) capital $z_{a,t}$.

$$\tilde{z}_t = \left[\rho_z\, (z_{i,t})^{\left(\frac{\xi-1}{\xi}\right)} + (1-\rho_z)(\nu_a z_{a,t})^{\left(\frac{\xi-1}{\xi}\right)}\right]^{\left(\frac{\xi}{\xi-1}\right)}.$$

where ν_a scales the productivity of adaptation capital in the aggregator, ρ_z is a weight parameter (governing the share of standard infrastructure in effective public capital), and ξ is the intratemporal elasticity of substitution between standard and adaptation capital. This formulation allows adaptation infrastructure to be either a substitute or a complement to regular infrastructure in production, depending on ξ (with the extreme case $\xi \to \infty$ implying perfect substitutability and $\xi \to 0$ implying Leontief complementarity).

Damage decomposition: Following Marto et al. (2018), the disaster intensity $D_{j,t}$ is conceptually decomposed into three components: (i) public capital destruction, (ii) private capital destruction, and (iii) transitory TFP losses. Let $\widetilde{D}_{j,t}$ denote the mitigated disaster intensity after accounting for adaptation's protective effect (as above). The model allocates this effective shock across different types of damage using fixed fractions. Specifically:

$$D_{z,j,t} = \kappa_z \widetilde{D}_{j,t},$$

$$D_{k,j,t} = \kappa_k\, \widetilde{D}_{j,t},$$

$$D_{A,j,t} = (1 - \kappa_z - \kappa_k)\widetilde{D}_{j,t}.$$

Here κ_z and κ_k (each between 0 and 1) govern how the overall disaster impact is split between permanent damage to public infrastructure vs. private capital. The remaining fraction $1 - \kappa_z - \kappa_k$ of the shock affects TFP temporarily. A higher κ_z (relative to κ_k) means the model attributes a greater portion of disaster damage to public capital losses (and less to private capital or TFP). In practice, these parameters are calibrated so that the model's implied output losses and capital destruction match observed post-disaster assessments (Marto et al., 2018). Damage to public and private infrastructure not only reduce those capital stocks immediately but also have persistent effects on productivity: infrastructure takes time to rebuild, and in the interim workers may face reduced efficiency due to damaged public services (power outages, impassable roads, etc.). Thus, even the "temporary" TFP loss component $D_{A,j,t}$ can last multiple periods until reconstruction is completed.

Public capital accumulation and investment efficiency: Standard and adaptation public capital accumulate according to:

$$z_{i,t} = (1 - \delta_{z_i})\, z_{i,t-1} + (1 - D_s)\, s_t\, i_{z_i,t}\,,$$
$$z_{a,t} = (1 - \delta_{z_a})\, z_{a,t-1} + (1 - D_s)\, s_t\, i_{z_a,t}\,,$$

where $i_{z_i,t}$ and $i_{z_a,t}$ are the flows of new investment in standard and adaptation infrastructure at time t, and $\delta_{z_i}, \delta_{z_a}$ are the depreciation rates. We assume adaptation capital is more durable than standard capital, so typically $\delta_{z_a} < \delta_{z_i}$ (reflecting the idea that resilient infrastructure better withstands wear-and-tear and disaster impacts). The term $s_t \in [0,1]$ is the public investment efficiency parameter, meaning that only fraction s_t of each dollar spent effectively adds to the capital stock. Any inefficiency $1 - s_t$ represents resources lost to bureaucracy, mismanagement, or other implementation frictions.

Recovery in the baseline (one-off disaster): In Marto et al. (2018), the economy is hit by a single major disaster at time $t = 0$. After this one-off shock, each disrupted variable x (for example, the public capital stock z_i, sectoral TFP A_j, the public investment efficiency s_t, or the real interest rate on external commercial debt r_{dc}) gradually recovers toward its pre-disaster steady-state value. The recovery path is modeled as a smooth convergence to the original level over time. Formally, if x_{pre} is the pre-disaster steady-state value and x_{ND} is the value of x immediately after the disaster (at $t = 0$), then for $t \geq 0$ one can write:

$$x_t \approx x_{\text{pre}} - (x_{\text{pre}} - x_{\text{ND}})e^{-\gamma^x t},$$

where $\gamma^x > 0$ governs the speed of recovery. A larger γ^x implies a faster return to normal. Thus, destroyed public and private capital are gradually rebuilt, TFP recovers as infrastructure is restored and workers return to full productivity, and any disaster-induced spike in borrowing costs or drop in efficiency dissipates over time.

2.2 Extensions: What Is New in the Augmented DIGNAD?

In our model, natural disasters are no longer a one-off event but rather a sequence of frequent shocks, reflecting the persistent exposure of disaster-prone economies to climate-related hazards. The economy is assumed to be hit every year by a shock of *average* magnitude equal to the country's Annual Average Loss. Each such shock exerts a permanent adverse effect on the level of GDP by destroying some public and private capital and by reducing productivity—just as in the one-off case—and public adaptation investment continues to mitigate these damages through the same channels described earlier. The key

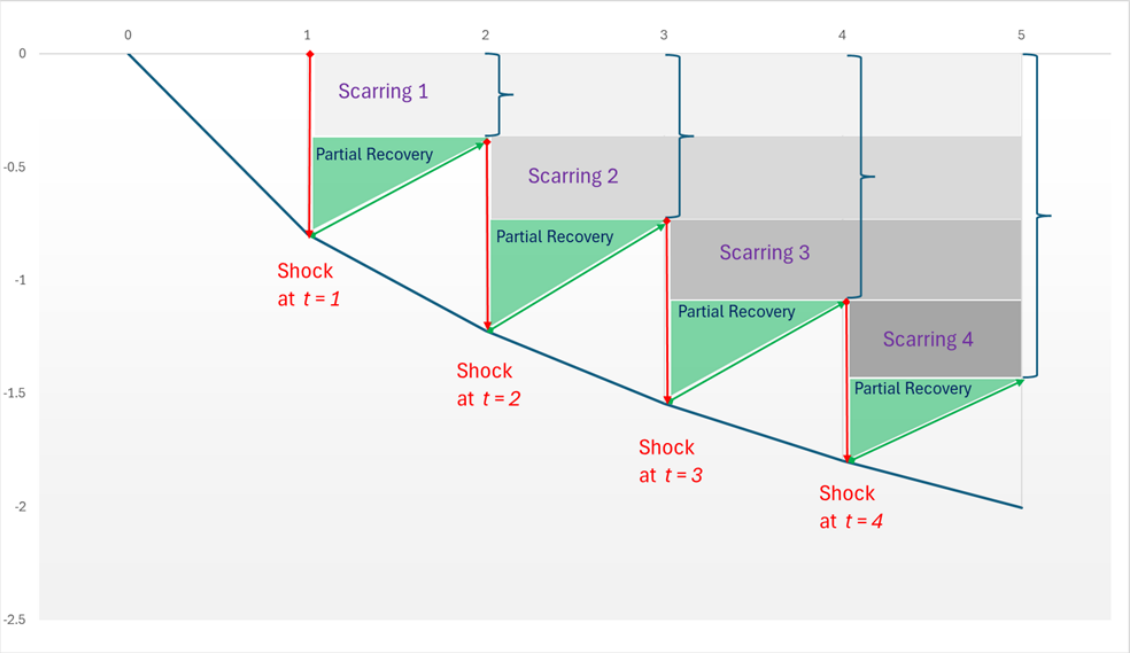
difference is that now shocks occur every period. For tractability, we calibrate the size of these recurring shocks to match the country's AAL. In practice, this means we impose a fixed disaster intensity $\bar{D}_j$ each year (for each sector j) such that the expected annual loss of output corresponds to the empirical AAL estimate. In a deterministic projection framework, we effectively set $D_{j,t} = \bar{D}_j$ for all t, treating $\bar{D}_j$ as the mean impact. In reality, actual disasters will vary in size year to year, but using AAL provides a convenient "average" shock for scenario analysis. The inherited mitigation term and damage decomposition then operate each year exactly as in the baseline model: adaptation capital reduces annual losses via the same structural formulas, and the fractions of damage going to TFP, public, and private capital (A_j, κ_z, κ_k) remain as in Marto et al. (2018).

Between-shock recovery dynamics: With a disaster striking every period, the economy never fully returns to its previous trajectory; rather, it exists in a constant cycle of partial recovery (Figure 1). After each annual shock, the affected variables begin to revert toward their prior levels, but the window before the next shock is limited. If reconstruction is incomplete or slow, key state variables (like public capital) remain below their no-disaster trend when the next shock hits, leading to an accumulating gap over time between the actual economy and its hypothetical no-disaster path. In our model, we allow different assumptions about the reconstruction response. Private capital is rebuilt endogenously by firms as they invest to replace destroyed capital (subject to adjustment costs). Public capital reconstruction, in contrast, can be more constrained. It might follow an exogenous process (e.g., a fixed multi-year rebuilding schedule) or be determined endogenously by government decisions (subject to fiscal and capacity limitations). We consider reconstruction scenarios in which the government mobilizes internal and external resources to partly rebuild lost infrastructure between shocks subject to implementation capacity. Even without a dedicated effort, the government will replace some lost capital simply to maintain essential infrastructure services, but inadequate capacity or funding can significantly slow this process.

Finally, there is a fundamental trade-off regarding adaptation investment. Resilient capital is costlier to build but suffers much less damage when disasters occur. Increasing $z_{a,t}$ (adaptation infrastructure) is expensive yet can greatly reduce future output losses. The model captures a trade-off between these long-term benefits and the debt implications of higher investment spending. Our model allows us to analyze this trade-off quantitatively: investing in resilience can substantially moderate the pace of public debt accumulation over time (also compared to standard infrastructure investment) by curbing disaster-related losses, whereas underinvestment in adaptation may lead to progressively worsening debt dynamics under repeated shocks.

Our extension essentially replaces the single-disaster experiment with recurring annual risk (calibrated to AAL) and adapts the recovery mechanism accordingly, while all other blocks of the model are inherited unchanged from Marto et al. (2018). In particular, household behavior and the broad structure of fiscal block remain the same as in the baseline DIGNAD framework.

Figure 1. Dynamics of Recurrent Natural Disaster Shocks under Partial Recovery
(Percent deviation from the no-disaster GDP path – numbers are illustrative)



Government sector

The government sector in our augmented DIGNAD model follows the same principles as in Marto et al. (2018). The government chooses how much to invest in standard vs. resilient infrastructure, faces an intertemporal budget constraint, and finances deficits via a mix of domestic debt, external debt (official and commercial), and support from donors (grants or concessional loans). Recurring natural disaster shocks translate into higher public investment needs (for reconstruction and adaptation) and lower revenue (due to output losses), which together tend to widen the fiscal deficit each period. The result is a rising public debt trajectory compared to a no-disaster baseline. By explicitly including the government's budget and debt dynamics, alongside the adaptation and disaster-loss mechanisms, we complete the description of our integrated AAL-DIGNAD framework. This will allow us to simulate how responses such as scaling up adaptation investment (compared to standard investment) can influence macroeconomic outcomes and debt dynamics under repeated disaster shocks.

Government's budget constraint: At each period t, the government's uses of funds must equal its sources of funds. The primary expenditure includes public investment (in both types of capital) and any transfers or recurrent spending, as well as debt service on outstanding debt. Sources of funds include new borrowing (domestic or external), tax revenues, any grants received, and withdrawals from the savings fund (if available). The consolidated budget identity can be written as:

$$P_{z_i,t} i_{z_i,t} + P_{z_a,t} i_{z_a,t} + \mathrm{Tr}_t + r_{t-1} b_{t-1} + r_d d_{t-1} + r_{dc,t-1} d_{c,t-1} + \Delta S_t = \Delta b_t + \Delta d_t + \Delta d_{c,t} + T_t + G_t + r_f S_{t-1}.$$

Intuitively, government spending on new infrastructure ($P_{z_i,t} i_{z_i,t} + P_{z_a,t} i_{z_a,t}$), plus any transfers or other spending Tr_t, plus interest payments on last period's domestic debt ($r_{t-1} b_{t-1}$), official (i.e., loans from

official lenders) external debt ($r_d d_{t-1}$), and commercial external debt ($r_{dc,t-1} d_{c,t-1}$), plus any addition to the savings fund ΔS_t (which is an outflow when the government sets aside resources), must be financed by the sum of domestic borrowing Δb_t, official external borrowing Δd_t, commercial external borrowing $\Delta d_{c,t}$, tax revenue T_t (e.g. from a consumption tax), grants G_t, and interest income $r_f S_{t-1}$ on existing savings.

Debt dynamics and financing mix: The model tracks domestic debt b_t (issued in local currency), official external debt d_t (at interest r_d), and external commercial debt $d_{c,t}$ (market loans at interest $r_{dc,t}$). New borrowing $\Delta b_t, \Delta d_t, \Delta d_{c,t}$ each period covers any financing shortfall. We impose a simple rule for the mix of domestic vs. external commercial borrowing: let $\upsilon \in [0,1]$ be a fixed share of external commercial financing in new non-official borrowing; then each period,

$$(1-\upsilon)\Delta b_t = \upsilon \Delta d_{c,t}\,, 0 \le \upsilon \le 1\,,$$

meaning a fraction $(1-\upsilon)$ of the additional financing comes from domestic bonds and υ from external commercial loans. Official external borrowing Δd_t and grants G_t are typically scenario-dependent (reflecting donor support) rather than chosen by the government.

Interest rates on debt are partly exogenous and partly endogenous. The rate r_d on official external debt is usually fixed (reflecting stable low interest from official creditors), and the world risk-free rate r_f is given. The external commercial borrowing rate $r_{dc,t}$, however, can include a sovereign risk premium that increases with the public external debt burden and can also spike due to disaster-related credit risk. We follow Marto et al. by specifying the external interest rate as:

$$r_{dc,t} = (1 + D_{r,t})[r_f + \eta(\frac{d_t + d_{c,t}}{y_t} - \bar{d/y})]\,,$$

where $D_{r,t}$ represents an exogenous disaster-induced premium (a temporary increase in borrowing cost when a major shock hits), η is a small elasticity capturing how the spread rises with the deviation of the external debt-to-GDP ratio $\frac{d_t + d_{c,t}}{y_t}$ from its long-run level $\bar{d/y}$, and r_f is the base world interest rate. In our AAL-based scenarios with moderate annual shocks, we assume $D_{r,t}$ is zero or very small (since recurring disasters of average size are anticipated by lenders). Still, this formulation allows us to examine worst-case situations where a particularly severe event could trigger a noticeable jump in borrowing costs (e.g., a scenario where losses from natural disasters would be time variant). Domestic debt b_t is assumed to carry a constant real interest rate r (for simplicity, as we abstract from monetary dynamics and assume ample domestic liquidity). We set r equal to the economy's long-run real interest rate.

To maintain debt sustainability in the very long run, an adjustment mechanism is typically needed. In the baseline DIGNAD, for example, a fiscal rule can gradually raise taxes (or cut spending) if debt rises too high. In our simulations, we focus on the medium-term horizon where recurrent disasters are driving up debt, and we assume the government primarily adjusts through borrowing and external support rather than immediate tax rate changes (reflecting the limited capacity of disaster-hit countries to raise taxes quickly). However, the model can incorporate such fiscal responses if needed for other scenarios not considered here. For instance, one could introduce a feedback rule for the consumption tax rate $\tau_{c,t}$ that slowly increases taxes when debt exceeds a target, as in Buffie et al. (2012) or Cantelmo et al. (2019).

3. Data and Calibration

Our augmented DIGNAD model is calibrated for a set of climate-vulnerable countries that experience frequent large natural disasters. We use a combination of cross-country data (covering 33 highly disaster-exposed economies – see table A1.1. in Annex 1 for a list of these countries) and information from disaster risk assessments in CDRI and adaptation cost studies, including estimates from Aligishiev et al. (2022) and the World Bank, to set key parameters.

3.1 Annual Average Loss (AAL) as a Disaster Shock Metric

A key innovation in our approach is to calibrate the recurring disaster shock in each country to its Annual Average Loss (AAL) – an estimate of the expected yearly loss from natural disasters in that country. The use of AAL allows us to incorporate probabilistic disaster risk information into a single model input. Specifically, we draw on AAL estimates from the Coalition for Disaster Resilient Infrastructure (CDRI) dataset, which provides AAL as a share of GDP or capital stock for multiple countries. The CDRI's AAL calculation is based on detailed hazard models and comprises three components (Cardona et al., 2023): (i) hazard – a probabilistic representation of all potential hazard events (from frequent to extremely rare) producing a Loss Exceedance Curve (LEC) for each country; (ii) exposure – the inventory and value of assets at risk; and (iii) vulnerability – relationships that translate hazard intensity into expected damage for each asset type. The AAL is essentially the integral of the LEC, combining all simulated events and their probabilities into one annualized expected loss figure (see Figure 2). CDRI provides AALs for six hazards – earthquakes, tsunamis, landslides, floods, tropical cyclones, and droughts – with the last four hazards incorporating projected climate change effects. Moreover, AALs are produced under multiple climate scenarios (lower, current climate, upper); in this paper we use results for current climate conditions.

Figure 2. Loss Exceedance Curve



Not all countries face macro-critical disaster risks. We identify 33 countries facing high climate-disaster vulnerability using both historical impact data (EM-DAT, 2000–2024) and modelled risk data (CDRI, 2023).[7] Countries are selected if they meet at least one of the following criteria:

- Population Affected: Average annual share of population impacted by disasters is in the top quartile globally (above ~1.6% per year), and the country experienced major disasters in at least 10% of years from 2000 to 2024 (per EM-DAT).
- Annual Losses as % of Capital Stock: AAL (as percent of total capital stock) is in the top quartile (greater than ~0.46% of capital stock per year), indicating potential damage to productive assets.
- Annual Losses as % of GDP: AAL (as percent of GDP) is in the top quartile (greater than ~1.7% of GDP per year), indicating large expected economic losses relative to the size of the economy.

[7] See EM-DAT - The international disaster database and GIRI & key figures | GIRI.

This selection yields a broad mix of countries. Some have relatively low AAL/GDP but high population exposure (e.g., Djibouti), others have extremely high AAL/GDP, but a smaller share of population affected (e.g., The Bahamas), and some have high AAL relative to GDP but low relative to their capital stock (e.g., Somalia, implying an under-developed capital base). Together these 33 economies cover a range of climate-related vulnerabilities (see Annex I for summary charts).

Comparing the magnitude of shocks in the original DIGNAD vs. our augmented version helps interpret the LEC and AAL. For example, Marto et al. (2018) calibrated the one-off disaster in Vanuatu (Cyclone Pam) as a ~60.8% of GDP output loss. In contrast, Vanuatu's AAL is about 1.8% of GDP. The AAL is much smaller because it represents an average annual loss (integrating over all events), whereas Cyclone Pam was an extreme tail event on the LEC (occurring perhaps once in several decades or more).

It should be noted that an AAL-based deterministic sequence is not equivalent to the full stochastic distribution of disaster events. AAL summarizes the expected annual loss implied by the loss-exceedance curve, but it does not preserve the higher moments of that distribution. In particular, a sequence of moderate annual losses can generate different dynamics from rare but very large events, especially when damages are nonlinear in event size and when large shocks produce persistent scarring. Thus, our baseline should be interpreted as a risk-adjusted expected-loss path, not as a replacement for tail-risk stress tests. This distinction is important because, for some countries, the AAL may be heavily influenced by low-probability, high-loss events. In those cases, spreading the expected loss evenly across years can understate nonlinear scarring from extreme events while overstating the extent to which continuous reconstruction offsets damages. For this reason, the baseline AAL calibration is best viewed as a central risk-adjusted expected-loss scenario. Alternative shock sequences, sensitivity across CDRI's lower, current, and upper climate scenarios, and Monte Carlo or other stochastic simulations would be useful robustness extensions but are beyond the scope of the present paper.

3.2 Calibrating Adaptation Investment and Recovery Parameters

A second key input to our model is the scale and effectiveness of adaptation (resilience) investments. Estimating the magnitude and returns to adaptation is challenging due to limited historical data and varying methodologies, but recent global studies provide useful benchmarks. For instance, the United Nations Environment Programme projects that global adaptation needs by 2035 could reach $300–360 billion per year (0.3% of world GDP), with one-third potentially financed by private sector contributions (UNEP, 2025). Another approach by Aligishiev et al. (2022) estimates that upgrading new and existing public assets to climate-resilient standards would cost about 0.2–0.4% of global GDP annually.[8]

Given the diversity of our country sample, we rely on cross-country estimates for consistency, adjusting for outliers as needed. Specifically, we draw on each country's estimated annual adaptation spending (as a percent of GDP) from Aligishiev et al. (2022), which offers a broad dataset. In cases where those estimates are missing or appear implausibly high/low, we supplement them with country-specific sources (e.g., World Bank Country Climate and Development Reports or other national studies). The heterogeneity in adaptation costs is striking: some low-income or small island countries require

[8] Throughout the paper, "adaptation investment" refers to the subset of adaptation spending that can be represented in DIGNAD as resilient public capital or disaster-risk-reduction infrastructure; this is narrower than economy-wide adaptation needs, which also include agriculture, water, health, ecosystem, and institutional measures.

adaptation investment of 2–3.5% of GDP per year (e.g., Lao PDR, Haiti, Somalia), whereas larger emerging markets often have lower proportional needs (1% of GDP or less) due to their bigger economies and more extensive existing infrastructure. Small island developing states (SIDS) and some low-income countries typically lie in between, facing moderate to high adaptation costs relative to GDP (Figure 3).

When evaluating the macroeconomic effects of adaptation investment, it is important to consider various parameters in the DIGNAD framework beyond just the scale of spending.[9] Key factors include:

- Returns to standard and adaptation infrastructure, captured by R_t^{zi} and R_t^{za} respectively
- Depreciation rate of standard and adaptation capital, captured by δ_{zi} and δ_{za} respectively
- Scaling parameter determining ability of adaptation capital to withstand ND: π_j
- Cost ratio of adaptation vs. standard investment (adaptation is % more expensive than standard): a_{za}

We calibrate the productivity returns to standard and adaptation capital at 30% and 50% respectively, and their depreciation rates at 7.5% and 3.0%, in line with Marto et al. (2018). However, we assume a relatively low-cost premium for adaptation, setting a_{za} to 1.03 (i.e., resilient infrastructure is about 3% more expensive than standard on average). Marto et al. (2018) assumed a much larger 25% premium (e.g., cyclone-resistant structures costing one-quarter more). Newer evidence suggests that incorporating resilience need not greatly inflate costs. However, if not done with proper planning and risk data, notably regarding spatial distribution of natural hazards, costs would rise dramatically (Hallegatte et al., 2019). Thus, our calibration implicitly assumes reasonably efficient, well-targeted adaptation investments. For simplicity, we also assume that prior to our simulations each country's stock of dedicated adaptation capital is zero (since historical data on resilient infrastructure is scarce), and we calibrate the public investment efficiency for each country to equal the value of the public investment efficiency variable developed at the IMF.[10]

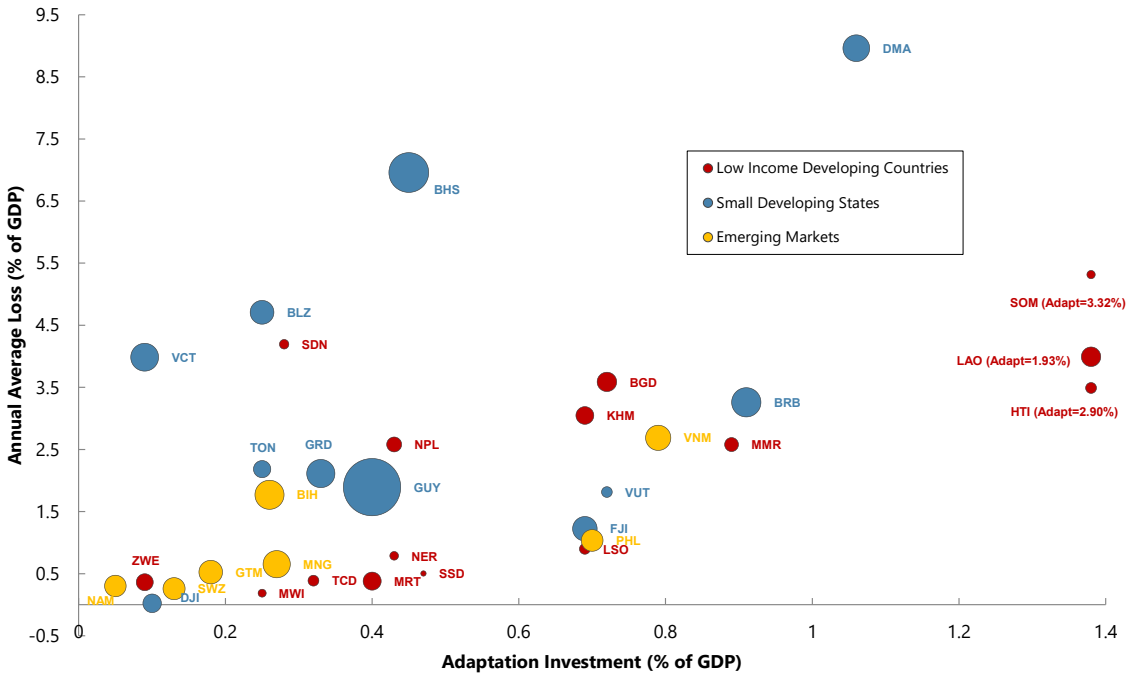
There is broad literature on the returns to adaptation investments, though most studies focus on specific sectors or projects. Scaling such micro-level benefits to an economy-wide impact is not straightforward. The World Bank's Country Climate and Development Reports (CCDRs) typically identify sectoral adaptation needs and interventions, then simulate those in macroeconomic models (like MANAGE or MFMod) to estimate overall impacts (see Abalo et al., 2025, and World Bank, 2025, for multi-country overviews). In our approach, we use CCDR findings primarily to validate our model's outputs — for example, checking that our simulated GDP impacts from adaptation align with those found in country-specific analyses (Section 4, Figure 5).

Finally, we need to specify the speed of recovery after each disaster shock under recurring conditions. Because shocks occur every year, the recovery process will inevitably be incomplete: the economy is in a constant state of "catch-up" rather than fully reverting to the no-disaster trend. For cross-country comparability, we adopt a simplifying common calibration under which about half of the damage is recovered one year after each event, roughly half the recovery duration assumed for a one-off event in Marto et al. (2018). This is not a country-specific empirical estimate; actual recovery may vary with hazard

[9] The DIGNAD user manual delves more deeply into these numbers (see Aligishiev et al, 2023).
[10] The investment efficiency parameter s_t captures how efficiently *overall* investment undertaken translates into a country's effective capital stock (see IMF, 2015).

type, institutional strength, financing availability, and implementation capacity. Country- and hazard-specific recovery calibration, including sensitivity to slower and faster recovery paths, is an important extension beyond the scope of the paper. This current choice implies that significant reconstruction takes place between annual shocks, but not enough to fully recover before the next disaster hits, thus causing persistent output scarring in the long run relative to a scenario without natural disasters.

Figure 3. Losses from Natural Disasters and Investment Adaptation Assumptions
(Bubble Size = GDP per capita in PPP | Color = Country Groupings)



Sources: CDRI, Countries' NDCs, World Bank WDI, Aligishiev et al. (2022) and staff calculations.

Note: For three countries with large adaptation investment estimates, lower calibration values are used (1% of GDP for Somalia and Haiti, and 0.5% for Lao PDR) for model convergence.

Table 1. Selected Parameters and Initial Values for Key Variables (in percent)
(Below values are common for all 33 countries, comparing to original Marto et al (2018) paper)

Definition	Version of DIGNAD Augmented	Original
Parameters determining main characteristics of infrastructure investment		
Cost ratio adaptation vs standard investment	3.0%	25.0%
Initial return on standard infrastructure investment	30.0%	30.0%
Initial return on adaptation infrastructure investment	50.0%	50.0%
Depreciation rate of public capital (standard infrastructure)	7.5%	7.5%
Depreciation rate of public capital (adaptation)	3.0%	3.0%
Scaling parameter determining ability of adaptation capital to withstand ND	14.0	25.0
Speed of recovery and reconstruction		
Speed of reconstruction of public standard infrastructure	50.0%	20.0%
Speed of recovery of public investment efficiency	50.0%	20.0%
Speed of recovery of tradable sector TFP	50.0%	20.0%
Speed of recovery of country risk premium	50.0%	20.0%
Firms		
Capital share in V.A. in non-traded goods sector	55.0%	40.0%
Capital share in V.A. in traded goods sector	40.0%	30.0%
Cost share of non-traded inputs in the production of private and public capital	50.0%	50.0%
Households		
Intertemporal elasticity of substitution of consumption	0.34	0.34
Portfolio adjustment costs parameter	100.0%	100.0%
Depreciation rate of private capital	5.0%	5.0%
Government		
Intra-temporal elasticity of substitution across public capital inputs	+∞	+∞
User fees for infrastructure services (% of recurrent costs)	0.0%	50%
Public debt risk premium parameter	0.0%	0.0%
Risk-free foreign real interest rate	4.0%	2.0%

Source: IMF staff calculations.

3.3 Other Calibration Issues

We run our model simulations from 2025 to 2058 (a 34-year horizon), treating 2024 as the base year and averages over the period 2025–30 as the initial state for most variables. Both the annual disaster shock (AAL) and the adaptation investment (in the adaptation scenario) are held constant as shares of GDP throughout the projection period. Holding AAL constant is reasonable given CDRI's forward-looking risk estimates (which implicitly consider current climate conditions over the coming decades). Keeping adaptation investment at a fixed GDP ratio is a simplification for illustrative purposes, consistent with some long-horizon analyses (e.g., the CCDRs often assume steady adaptation spending over ~30 years).

Table 2. Initial values for key variables for selected countries

Definition	PHL	MRT	BRB	VUT 1/
Public Infrastructure Investment to GDP (standard capital)	4.4%	2.5%	1.3%	1.3%
Consumption tax (VAT) rate	12.0%	16.0%	17.5%	12.5%
Public Domestic Debt to GDP	43.4%	8.1%	54.4%	7.7%
Public External Official Debt to GDP	7.9%	39.7%	28.5%	12.9%
Public External Commercial Debt to GDP	8.4%	0.0%	6.7%	5.0%
Private External Debt to GDP	11.0%	1.2%	15.4%	0.0%
Real Interest Rate on Public Domestic Debt	2.8%	2.7%	1.5%	5.0%
Real Interest Rate on Public External Debt	2.1%	-1.0%	1.5%	1.5%
Grants to GDP	0.0%	2.6%	0.2%	11.2%
Remittances to GDP	9.1%	0.9%	1.4%	3.5%
Imports to GDP	32.2%	42.5%	36.3%	54.0%
Government Savings to GDP	2.8%	6.4%	3.4%	0.0%
Value Added in NT Sector	62.0%	43.2%	74.1%	48.0%
Ratio of Non-Savers to Savers	0.2%	2.7%	0.2%	12.7%

Source: IMF staff calculations.
1/ From Marto et al. (2018) paper presenting the original DIGNAD model.

Notes: Table entries are DIGNAD calibration inputs and not official 2024 numbers. Sources include IMF databases (such as WEO), CDRI, PWT11, World Bank WDI, World Bank Findex, International Debt Statistics. Variables linked to legislation or infrequently updated surveys use the latest available observation, while macroeconomic and debt-related variables are calibrated using averages. Missing observations are filled using secondary sources, country matching, development-group medians, or zero values where conceptually appropriate. The ratio of non-savers to savers is calibrated using financial account ownership data. Specifically, the model uses $a_ratio = (1 - AccountOwn)/AccountOwn$, where AccountOwn is the share of adults with an account at a financial institution or mobile-money-service provider. World Bank Global Findex is the main source; missing values for some small states are filled using regional financial-inclusion surveys. The model's non-tradable sector is proxied using services value added from WDI. This is a practical cross-country calibration choice, but it is not conceptually identical to non-tradables. In economies with large export-oriented service activities (e.g., the Philippines), some services are tradable. We therefore interpret the NT share as a model calibration proxy rather than a literal measure of non-tradable production.

In Sections 3.1 and 3.2 above, we detailed the two main scenario inputs in our model (disaster shocks via AAL, and adaptation investment assumptions) and our calibration of key related parameters. Table 1 summarizes selected parameter values that are common across the 33 countries, which largely mirror those in Marto et al. (2018) except where noted (such as the adaptation cost premium). We do not discuss those common parameters at length, as they do not drive differences in results. Table 2 provides an overview of important country-specific initial values for three illustrative cases (Philippines, Mauritania, Barbados) and compares them to those for Vanuatu in Marto et al. (2018).[11] Some initial conditions (like public investment efficiency or real interest rates on debt) will significantly influence outcomes across countries, while others (such as initial grant inflows or remittance levels) have limited bearing on the dynamic results in our simulations.

The simulations in the remainder of the paper are deterministic transition experiments solved under perfect foresight, conditional on the specified sequence of disaster shocks and financing rules. In the risk-adjusted baseline, each economy is initialized at its 2025–2030 macroeconomic state and is then exposed, from 2025 onward, to an annual disaster shock calibrated to the country-specific AAL. The shock affects public capital, private capital, and TFP through the same damage channels as in the

[11] Results for other countries are available upon request.

original DIGNAD model, while the recovery mechanism governs the extent to which damaged variables revert before the next annual shock arrives. The model is therefore not solved as a stochastic dynamic programming problem in the baseline exercises. Rather, the recurrence of shocks is represented as a pre-specified deterministic sequence of annual shocks, and the economy's transition path is computed under the fiscal closure and financing assumptions described in the text. Alternative scenarios are generated by changing one policy block at a time—standard public investment, adaptation investment, or the AAL path—while holding all other calibration choices fixed. This design allows the paper to isolate the macro-fiscal effect of recurrent natural disasters and the marginal costs/benefits of adaptation investment. It also implies a clear limitation: the baseline does not capture precautionary behavior or stochastic steady-state effects arising from agents' expectations about the full distribution of rare disaster events. Concretely, the simulated private investment responses do not incorporate precautionary behavior, and financing and debt paths remain conditional on the announced deterministic shock sequence and financing rules; the paper does not establish robustness to alternative expectations assumptions.

4. Results and Analytical Implications

We first present an overview of model results across the 33 high-disaster-risk countries in our sample, then discuss specific case studies (the Philippines, Mauritania, and Barbados) to highlight diverse country experiences and trade-offs. In all scenarios, we compare a risk-adjusted baseline (with recurring disasters but no additional investment) against alternative scenarios.

4.1. Cross-Country Baseline and Adaptation Scenarios

Our risk-adjusted baseline scenario incorporates each country's AAL as an annual disaster shock, under current levels of resilience investment. Unsurprisingly, these recurring natural disasters have a cumulative negative effect on GDP and public debt. Over a 30-plus-year horizon, the affected economies experience progressive reduction in output and rising public indebtedness relative to a no-disaster counterfactual. This highlights the critical importance of integrating climate-related disaster risks into macro-fiscal analysis and planning; otherwise, standard projections could suffer from an upward bias in growth and debt sustainability assessments (consistent with concerns raised by Kahn et al., 2021 in the context of the growth impact of slow-moving temperature increases). Relative to a one-off disaster stress test, the recurring-loss framework highlights a different macro-fiscal mechanism. The issue is not only the immediate financing need after a large event, but the gradual erosion of the output and revenue base as expected losses recur over time. This can lead to persistent financing pressures and debt drift even in the absence of a single catastrophic shock. The framework also sharpens the distinction between standard and resilient investment: under recurring shocks, non-resilient capital is repeatedly exposed to damage, while resilient capital yields benefits through avoided future losses.

Next, we examine two alternatives: an Adaptation Investment scenario and a Standard Public Investment scenario. In both cases, the government scales up annual public investment by a fixed amount (specific to each country, based on Section 3.2 estimates), financed entirely by additional borrowing (split evenly

between domestic and external commercial debt for illustrative purposes).[12] The only difference is the nature of the investment: in one case, funds are devoted to adaptation (resilient infrastructure), while in the other they are spent on conventional infrastructure projects that do not enhance disaster resilience. By comparing these scenarios, we can isolate the macroeconomic benefit of investing in resilience as opposed to simply increasing public capital.

Our simulations show that investing in resilience yields substantial net benefits in terms of output, but compared to the risk-adjusted baseline, there is a fiscal price to pay (Figure 4). When comparing to a scenario with the same spending on standard infrastructure, the adaptation scenario is better across both output and debt outcomes (Figures 6 and 7). Across our sample of 33 countries, the Adaptation Investment scenario results in materially smaller cumulative GDP losses from recurring disasters relative to the baseline. On average, about one-quarter of the long-run output loss is averted compared to making no extra adaptation efforts (median recovery of about 25% of the baseline output gap over 34 years). The public debt path, however, deteriorates moderately due to the additional borrowing required for adaptation. In the median country, public debt ends up roughly 5–6 percentage points of GDP higher by the end of the projection horizon, relative to the risk-adjusted baseline without adaptation. This outcome underscores a core trade-off: resilience investment can "buy down" future disaster losses, but it must be financed, often increasing debt in the short-to-medium run. These results also have implications for debt sustainability analysis. A climate-blind baseline may understate debt risks by excluding recurring disaster losses, while debt-financed adaptation may raise projected debt before the benefits of avoided losses materialize. Risk-adjusted expected-loss baselines can therefore help capture both the fiscal cost of adaptation and the fiscal benefit of avoided disaster damage. More generally, borrowing-financed adaptation can improve resilience while still leaving debt vulnerabilities unresolved, especially where fiscal space is limited.

In contrast, the Standard Investment scenario (allocating the same amount of borrowing to standard public capital) provides much smaller protection against disaster impacts. While conventional infrastructure raises productive capacity, it remains more vulnerable to destruction, so its benefits are largely eroded by the recurring shocks. As a result, GDP losses remain closer to the baseline path. Certain countries even suffer higher net losses if additional standard capital is repeatedly destroyed. Yet the debt trajectory in the standard investment scenario is still higher than the baseline (due to the extra borrowing). Taken together, these results demonstrate that adaptation yields a far higher "return" in terms of avoided economic damage per dollar of debt incurred, relative to just scaling up non-resilient investment. In short, resilience is a better use of scarce public resources in disaster-prone settings.

It is important to note the considerable heterogeneity in outcomes across countries. The relative benefits of adaptation versus standard capital vary depending on countries' structural characteristics like initial public capital stocks, public investment efficiency, fiscal space, and the relative magnitude of the AAL

[12] Unlike in the original DIGNAD model, we do not investigate alternative financing scenarios (e.g., savings funds, tax increases) and instead focus on borrowing to isolate the output-debt trade-off. In practice, adaptation and residual-disaster-risk financing may be organized through a risk-layered architecture that combines fiscal buffers, contingent credit, concessional resources and grants, disaster-risk insurance and other risk-transfer instruments, and catastrophe-contingent debt instruments. We do not model this financing architecture here, nor do we test alternative domestic-versus-external financing mixes or borrowing-cost assumptions. These instruments and choices would affect both the feasibility and the debt-sustainability implications of adaptation investment and the financing of residual disaster risk.

shock. For example, countries with greater fiscal space and investment efficiency see larger long-term gains from adaptation spending, since they can implement resilience projects effectively and absorb higher debt increases without destabilizing the economy. By contrast, highly vulnerable countries with limited fiscal headroom or significant absorptive capacity constraints (e.g., certain low-income countries and small states) may experience more modest gains from adaptation and a higher risk that increased borrowing for investment leads to debt sustainability concerns. Overall, adaptation is not a mechanical solution: its payoff depends on hazard targeting, implementation capacity, financing terms, and public investment efficiency. In low-capacity or fragile settings, high needs can coincide with weak capacity and tight fiscal constraints, limiting realized gains.

Figure 4 depicts this heterogeneity of impacts. Countries like Mongolia, Mauritania, the Philippines, Fiji, and Vanuatu rebound strongly from GDP losses, recovering over 50 percent. The first three are emerging markets, which may suggest they have better public financial management systems and more resources to handle these shocks. On the other hand, Cambodia, Sudan, Lao PDR, and Zimbabwe—classified as low-income countries (LICs)—struggle, with less than 10 percent GDP recovery, supporting the idea that poorer nations face greater challenges responding to natural disaster shocks. Interestingly, countries such as Belize and the Bahamas also show very low recovery rates, despite not fitting this pattern. Some nations appear to avoid the typical trade-off. For example, despite its large AAL, Dominica's high public capital stock and strong public investment efficiency allow high adaptation spending to slightly improve debt outcomes while reducing GDP losses by 25 percent. In Djibouti and Malawi, yearly adaptation investments exceed the AAL, resulting in growth that surpasses the scenario without natural disasters, even though debt levels still worsen. Niger stands out because GDP is lower in the adaptation scenario than in the risk-adjusted baseline. This is likely because the assumed adaptation package does not generate sufficient protective or productivity gains to offset its financing cost and implementation frictions.

The model should not be read as implying that adaptation infrastructure is intrinsically more productive than standard infrastructure in all states of the world. In normal times, a road or bridge may provide similar services whether built to standard or resilient specifications. The economic advantage of adaptation arises because resilient capital reduces expected future losses, depreciates less under climate stress, and preserves the productivity of private and public capital when disasters occur. However, it is still more expensive than investment in standard infrastructure. The case for adaptation is therefore state-contingent: it is strongest where disaster exposure is high, public investment is efficient, financing costs are manageable, and resilient assets are well targeted to the relevant hazards. This is also why some countries may see limited gains from adaptation despite bearing the fiscal cost.

These results are consistent with evidence from the World Bank's CCDRs. A recent overview of CCDR findings by the World Bank indicates that, when controlling for Small Island Developing State (SIDS) status (which is characterized by unique climate exposures and income profiles), higher-income countries tend to exhibit greater resilience to climate change. Specifically, each additional $1,000 in GDP per capita is associated with approximately a 0.26 (0.02–0.50) percentage point lower GDP loss due to climate impacts (World Bank, 2025, p.6). Figure 5, which is sourced from World Bank (2025), illustrates the estimated returns on adaptation investments. Across the countries examined, proactive adaptation measures are projected to reduce climate-induced GDP losses by about 4.2 percentage points on average. Notably, in countries such as Benin, the Dominican Republic, and Grenada, these adaptation efforts could more than halve the projected mid-century impacts of climate change on GDP.

To illustrate these points concretely, we turn to case studies of three countries from our sample (the Philippines, Mauritania, and Barbados). Each represents a different type of climate-vulnerable economy, demonstrating how country-specific factors influence the macro-impact of recurring disasters and the efficacy of adaptation policies. Importantly, the case-study simulations are illustrative model-based scenarios. They are not official IMF forecasts, country-team baselines, or program projections, and they do not incorporate all ongoing policy commitments, fiscal consolidation plans, or program targets.

Before turning to the three case studies, it is important to note that the financing implications of recurrent disasters differ qualitatively from those of one-off shocks. A large one-off disaster mainly creates an immediate liquidity need as the government must finance emergency relief, reconstruction, and temporary revenue losses until activity recovers. Recurrent disasters create a deeper solvency challenge. If annual average losses repeatedly damage the capital stock and lead to permanently lower levels of output, the debt ratio can deteriorate even in the absence of a single catastrophic event. In such an environment, the model indicates that short-term liquidity support alone does not address the longer-term debt dynamics. Financing instruments that reduce the cost of long-term adaptation investment or share risk across states of nature, may limit disaster-related borrowing from snowballing into debt problems.

Figure 4. Change in Debt/GDP and Recovered GDP Loss owing to Adaptation Investment

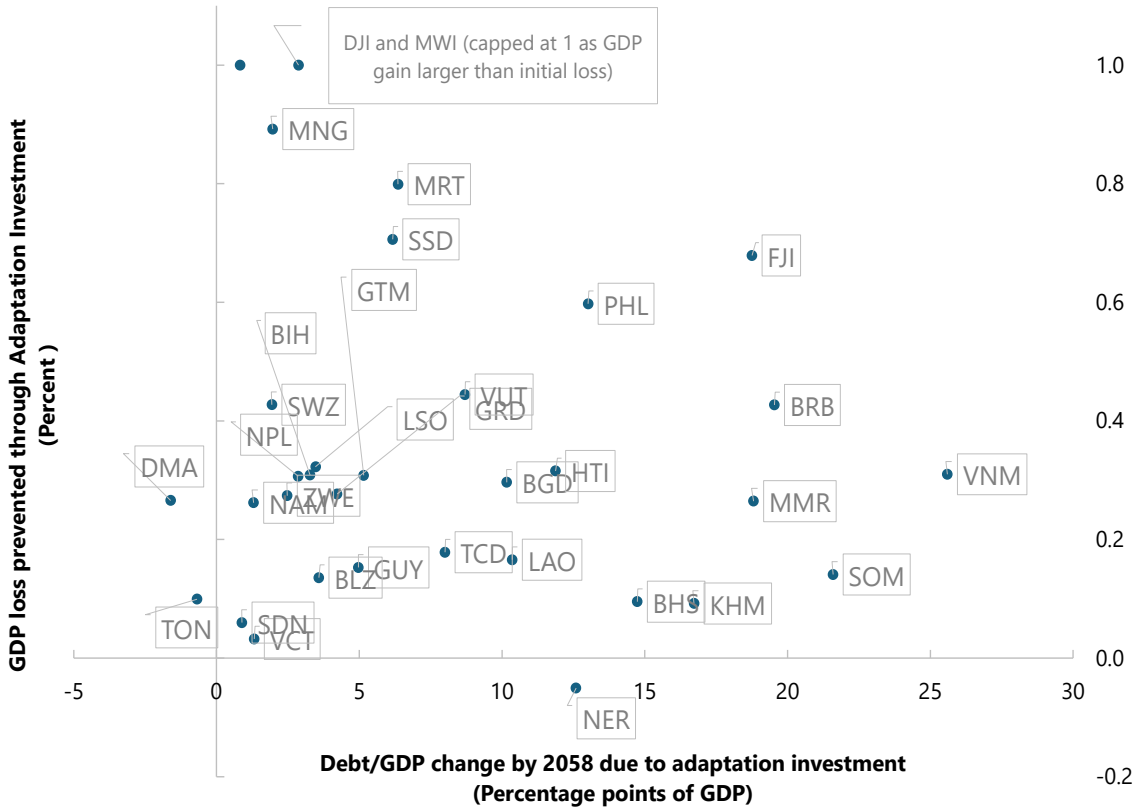
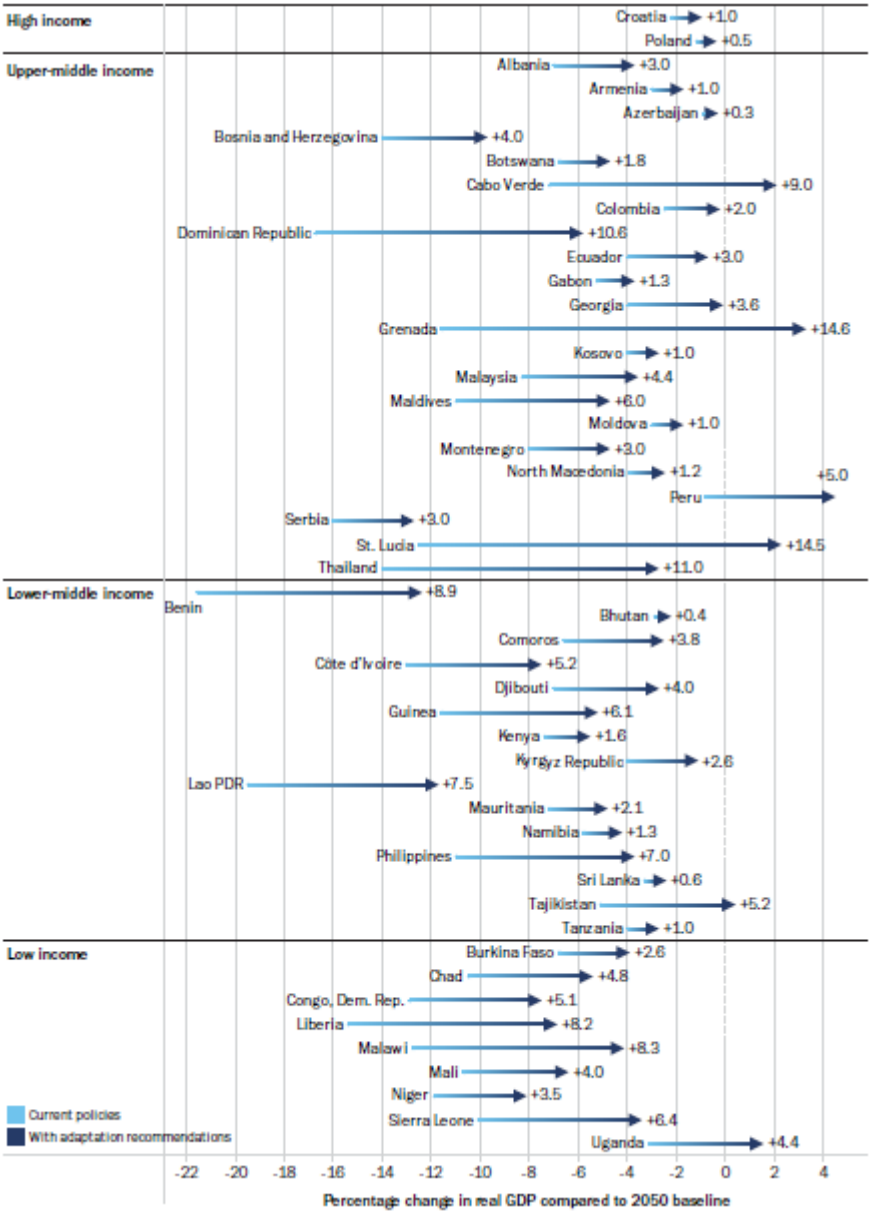
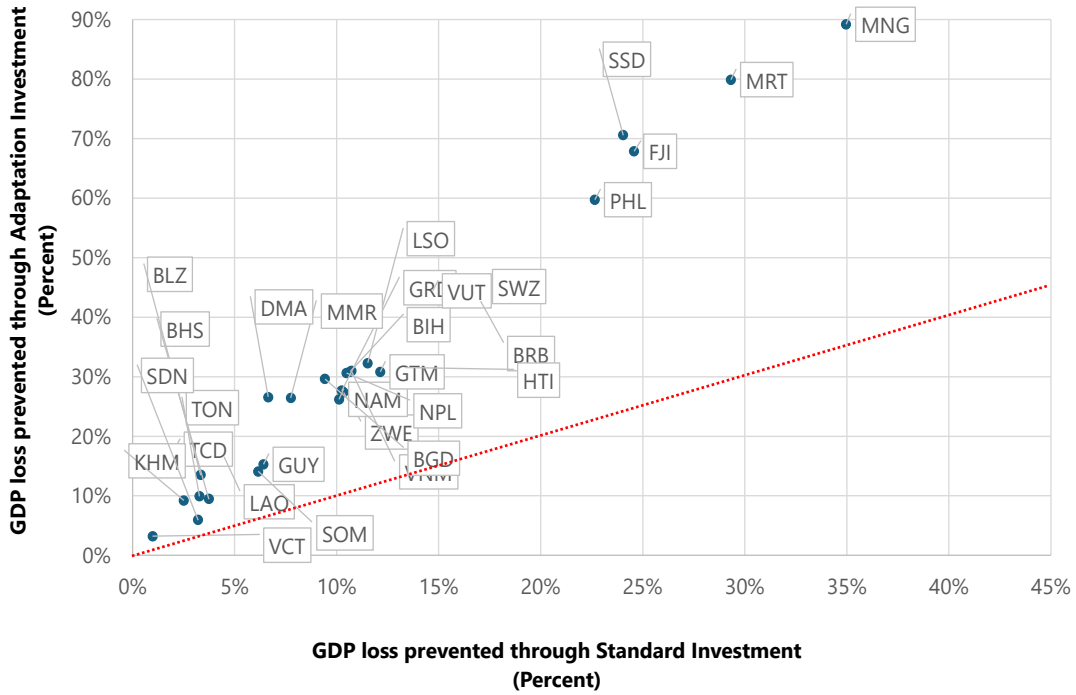


Figure 5. Growth Benefits from Adaptation Investment in World Bank CCDRs



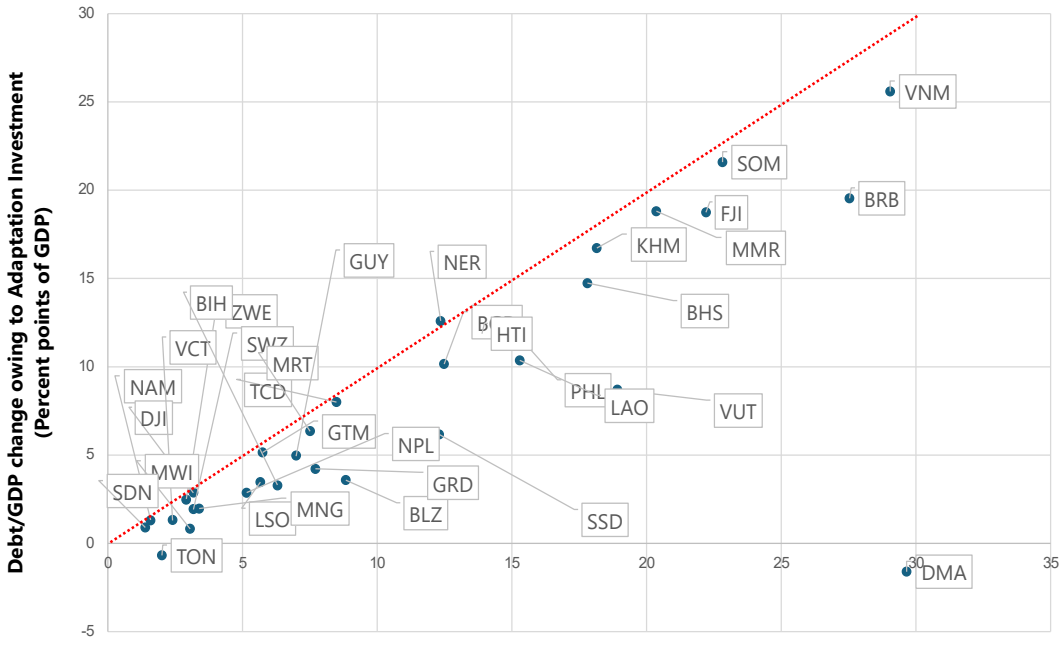
Source: Figure 7 in World Bank (2025). The figure shows the GDP delta between a climate scenario without adaptation investment and one with adaptation investment from World Bank CCDRs, by 2050. All countries shown in the figure avoid some GDP losses via adaptation investments.

Figure 6. GDP Effects of Standard Investment vs. Adaptation Investment



Note: DJI and MWI not shown as GDP loss prevented under adaptation scenario is larger than 100%. NER not shown as growth under both adaptation and standard investment scenarios deteriorated.

Figure 7. Debt Impact of Standard Investment vs. Adaptation Investment



4.2. Country Case Studies: Philippines, Mauritania, and Barbados

In this section we take a closer look at the output of the augmented DIGNAD model for three specific countries. Table 3 compares results between the augmented DIGNAD model developed in this paper with similar exercises developed at the IMF and WB. Figures 8, 9 and 10 then show the key outputs including dynamics of the current model.

Table 3. DIGNAD and CCDR[1/] results for Philippines, Mauritania and Barbados

	Size of ND shock (% of GDP) 2/	Annual Adaptation Inv. (% of GDP)	Adaptation Inv. Growth Gain (% 'clawed back')	
PHILIPPINES 3/				
Original DIGNAD	4.30%	1.2% for 5 yrs	1.4%	Contemporaneous GDP loss
Augmented DIGNAD	1.03%	0.7% during 2024-58	60%	GDP loss recovery
World Bank CCDR	11.00%	0.7% during 2020-50	65%	GDP loss recovery
MAURITANIA 4/				
Original DIGNAD	11.50%	2% for 5 yrs	7%	Contemporaneous GDP loss
Augmented DIGNAD	0.38%	0.4% during 2024-58	80%	GDP loss recovery
World Bank CCDR	7.20%	not specified	29%	GDP loss recovery
BARBADOS				
Original DIGNAD 5/	16.80%	1.0% for 5 yrs	5%	Contemporaneous GDP loss
Augmented DIGNAD	3.26%	0.9% during 2024-58	43%	GDP loss recovery

1/ CCDRs adaptation investment also covers slow-moving impacts of climate change.
2/ For CCDRs this is GDP losses by 2050 relative to no CC baseline. CCDR results are derived from data shown in Figure 5.
3/ Original DIGNAD from IMF (2023a), CCDR from World Bank (2022a).
4/ Original DIGNAD from IMF (2023c), CCDR from World Bank (2022b).
5/ As detailed in IMF (2023b).

4.2.1 Philippines (Emerging Market Case)

The Philippines is consistently ranked among the most at-risk countries globally for extreme events, with historically high exposure to typhoons.[13] On average, the country experiences the most intense Category-5 typhoons twice a year, with lower-intensity typhoons occurring more frequently. For the Philippines, we use an AAL shock of 1.03% of GDP from CDRI distributed across a large economy with some resilience due to diversification. The Philippines' initial conditions include moderate public debt (~60% of GDP), a diversified economy with significant tradable and export-oriented service activities, and ongoing efforts to strengthen public investment efficiency through PFM reforms, though with pockets of vulnerability (e.g., high disaster exposure in certain regions and heavy reliance on agriculture as a source of employment).

Under recurring disaster shocks, our model suggests that Philippine real GDP would be about 5 percent lower by 2058 than in the no-disaster scenario, reflecting the cumulative effects of annual natural-disaster damage. Over 2025–2058, the Philippines' public debt-to-GDP ratio rises above its no-disaster trajectory by 5 percentage points, despite relatively strong underlying growth. The adaptation scenario which scales up public investment by 0.7% of GDP annually significantly mitigates output losses.[14] According to our simulations, 60% of the output loss could be avoided, but this comes with a cost of an extra 14.5

[13] See the World Bank's CCDR for a discussion of the Philippines' case (World Bank, 2022a).
[14] For specifics on optimal interventions such as flood defense or climate-smart infrastructure at the sectoral level, please refer to the World Bank's CCDR.

percentage points of debt-to-GDP by 2058. The size of avoided income loss is similar to the World Bank's CCDR result of 65% over a 30-year period.

Crucially, these results emphasize that for an EM with good capacity, adaptation can be a high-return investment (including when compared to standard infrastructure investment). The adaptation scenario for the Philippines yields a more favorable growth trajectory, providing higher GDP (hence higher revenues and potentially more fiscal space in the long run) than the risk-adjusted baseline. In other words, a more resilient capital stock reduces yearly disaster damage despite higher debt levels (which are, in any case, lower than under a scenario with standard infrastructure investment of the same size). Our model would suggest that adaptation investment – if effectively implemented – will pay for extra growth and fiscal dividends over time. Nonetheless, debt would increase further. These model-based findings are broadly consistent, at a high level, with the Philippines' Disaster Risk Financing and Insurance (DRFI) Strategy, which emphasizes proactive resilience and risk layering across fiscal buffers, contingent credit, insurance and other risk-transfer instruments, and concessional finance.

In all three scenarios, private investment decreases as growth is hit (and does not recover to the no-disaster counterfactual in any of the three scenarios), through destruction of capital and reduction of TFP, both of which makes new capital less productive. However, this effect is muted in the scenarios where public investment is increased as that accumulation of public capital increases the marginal product of total capital, inducing more private capital investment. The same reasoning applies as to why under the adaptation investment scenario, private capital holds up better. This positive effect needs to be set against the crowding out effect of the increase in public capital, however, as less financing will be available for private investment.

Finally, the size of the shock in the one-off DIGNAD simulation (IMF, 2023a) is set higher at 4.3% of GDP, which is calibrated on a Category 5 typhoon from 2013 with losses estimated by EM-DAT. After 4 years of adaptation investment worth an annual 1.2 percent of GDP, the loss in the year of the disaster is found to be 1.4 percentage points of GDP less than in the scenario without those 4 years of investment.

4.2.2. Mauritania (Low-Income Developing Country Case)

Mauritania provides a starkly different context: a low-income country (LIC) in the Sahel, with a mix of climate threats (droughts, floods) and limited economic diversification. It doesn't face the dramatic cyclones of a tropical island, but it endures chronic aridity and periodic severe droughts and flash floods that threaten agriculture and infrastructure. Using CDRI's modelled data, Mauritania's recurrent disaster risk (AAL) is 0.4% of GDP affecting its subsistence sector and rural livelihoods. Though these events often don't attract the global attention of a hurricane, their cumulative impact is substantial.

In the risk-adjusted baseline scenario, Mauritania's recurring shocks lead to an increasing divergence from its no-disaster GDP path, reflecting the scarring typical of the augmented DIGNAD model. Even though the country experiences some recovery each year (e.g., regains some lost capital and TFP as rain returns or infrastructure is repaired), the next shock hits before full normalcy is achieved, gradually reducing the GDP level. Over decades, this yields a lower level of GDP than a climate-stable scenario (i.e., recurrent disasters impose a long-lasting output loss). Concurrently, recurring shocks lead to sustained fiscal deficits – lost revenues and increased spending on relief and minimal repairs – causing public debt to grow steadily beyond the no-shock case. Summarizing, compared to a no recurring disaster set up, by 2058 GDP is 1.3 percent lower and the debt-to-GDP ratio is 3 percentage points higher.

In the adaptation scenario, we simulate an additional 0.4% of GDP per year devoted to resilience (e.g., improved water management, drought-resistant infrastructure, flood defenses – reflective of Mauritania's adaptation needs identified in World Bank, 2022b). This significantly reduces the recurring GDP losses. In fact, in Mauritania's calibration, our model suggests adaptation investments could substantially cut GDP losses (80% reduction in disaster-induced GDP decreases), aligning with empirical evidence that well-targeted resilience can drastically mitigate losses. However, financing adaptation investment of 0.4% of GDP annually via debt implies a notable debt burden increase of about 10 percentage points of GDP. Therefore, the adaptation vs. no adaptation is clear: while growth improves, debt increases compared to the risk-adjusted baseline. The advantages of investing in adaptation become more apparent when comparing debt and growth paths to those under standard infrastructure investment of the same size, which leads to a higher debt-to-GDP ratio and lower GDP. In our simulations, the public-debt-to-GDP ratio in Mauritania still rises under all scenarios due to recurring deficits; adaptation somewhat moderates that rise by preserving output and limiting some shock costs, but to fully stabilize or reduce debt would likely require either external support (like concessional financing or grants) or a fiscal adjustment outside the scenarios considered (e.g., domestic revenue mobilization).

4.2.3. Barbados (Small State Case)

Barbados typifies a small island developing state (SIDS) with high exposure to tropical cyclones and storm surges. As a higher-income small economy, it has somewhat better infrastructure than average LIDCs but remains highly vulnerable due to its size and concentration of assets (e.g., tourism infrastructure near the coastline) – factors that can lead to outsized impacts when disasters strike. Barbados has implemented notable climate resilience strategies in recent years, including a Resilience and Sustainability Facility program with the IMF focusing on climate adaptation and macro resilience. In our analysis, we consider a scenario where Barbados (absent external support beyond baseline) faces recurring moderate storms each year.

For Barbados, we calibrate the AAL shock mainly via a combination of intense but infrequent tropical cyclones and some coastal events (drawing on hazard models), settling on annual disaster damage of 3.26 percent of GDP. We calibrate initial conditions for Barbados reflecting its 2025–30 state: high public debt, moderate growth potential, and public investment efficiency as measured by available cross-country indicators used in the calibration.

Under recurring shock baseline, Barbados experiences a slow, persistent erosion of GDP relative to a no-disaster trend, akin to a small permanent shock each year with partial recovery in between – in line with the expectations of the augmented DIGNAD model. Public debt in Barbados remains high in all scenarios, reflecting pre-existing debt and limited growth potential, but recurring shocks cause debt to drift upward. In the adaptation scenario, we simulate Barbados investing an additional 0.9% of GDP per year in resilience. Given Barbados's relatively high per capita income and institutional capacity, one could expect a relatively effective use of adaptation dollars. Our results for Barbados suggest that adaptation leads to moderate improvement in protecting GDP, reclaiming 43 percent of GDP loss compared to the recurring natural disaster baseline scenario. This is lower than our results for the Philippines and Mauritania, as for Barbados, being a small open economy with limited diversification, the direct growth payoff from adaptation is somewhat less pronounced. Still, adaptation benefits show up in a lower needed reconstruction spending each period, allowing Barbados's debt trajectory to remain more contained than under the standard infrastructure investment scenario: by 2058, the adaptation scenario's debt ratio is a few percentage points below the standard infrastructure investment scenario but still higher than the risk-

adjusted baseline. We note that, because Barbados already had a very high starting debt, whether adaptation financed by further debt is feasible depends on outside support – but our model's comparative static results indicate that if additional funds can be secured under alternative financing assumptions (preferably concessional or grant-based), the model indicates positive macro returns in terms of resilience. This aligns with the view that climate-vulnerable island states can benefit hugely from upfront resilience spending, but financing these without outside help is tricky given their debt constraints.

These results also underscore that adaptation investment alone may not be sufficient to preserve debt sustainability when financed through borrowing; debt outcomes may also depend on the availability of complementary fiscal measures, concessional financing, grants, or risk-sharing instruments.

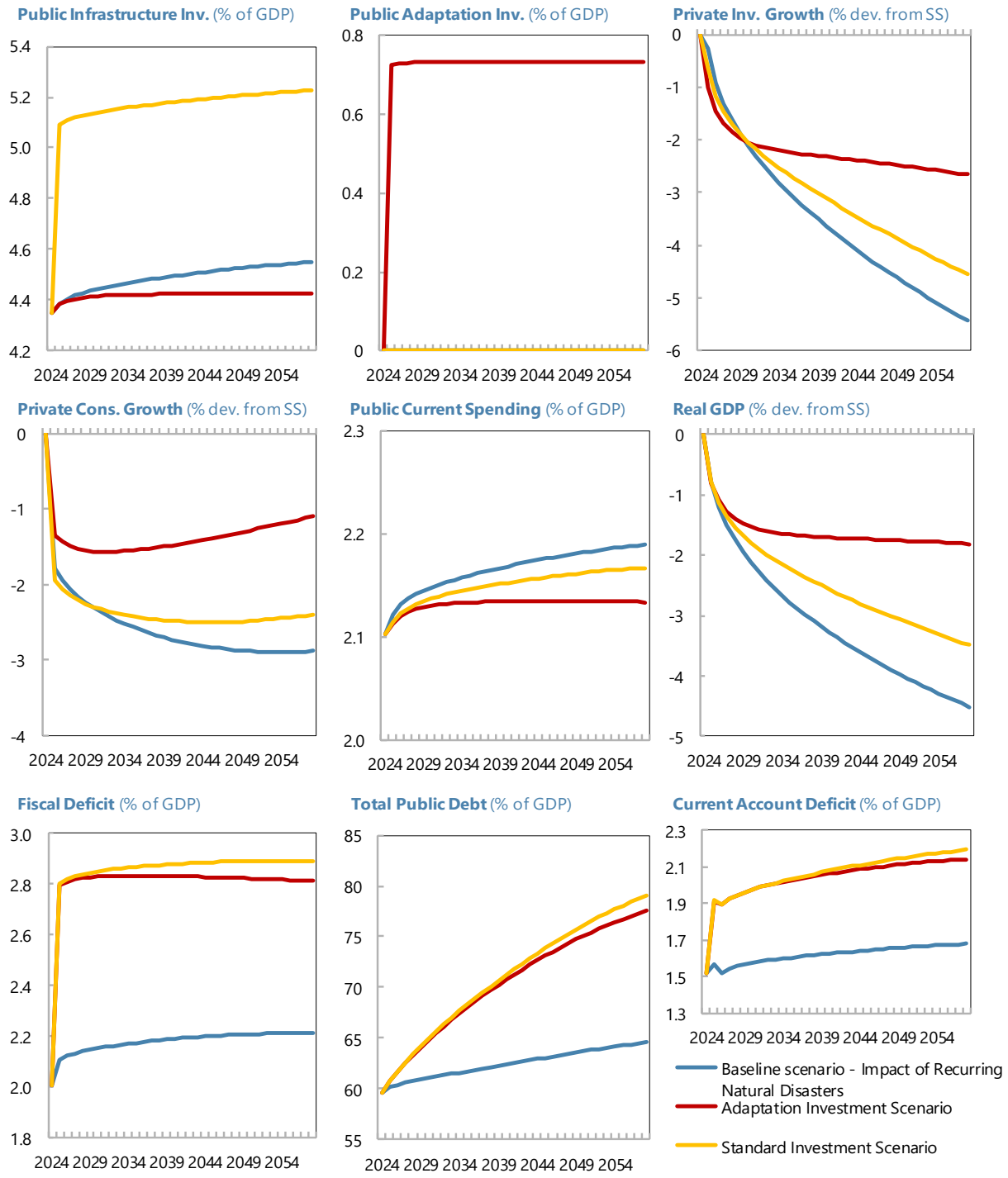
5. Conclusions

We developed an integrated macro-fiscal modeling approach to account for the recurring nature of natural disasters in climate-vulnerable economies. By extending the DIGNAD model to include annual disaster risk through AAL-based shocks, we generated risk-adjusted growth and debt projections that differ markedly from standard climate-blind baselines. Our results indicate that failing to incorporate recurring disaster risks can lead to systematically biased projections, underestimating the headwinds to growth and the pace of debt accumulation in disaster-prone countries. This is the central distinction from traditional one-off DIGNAD applications: recurrent disaster risk changes the projections itself, rather than only generating deviations around a climate-naïve path. The main analytical implications therefore complement, rather than overturn, those from standard DIGNAD: resilience investment remains important, but its benefits should be assessed against a path that already reflects expected recurring losses.

A key finding is that adaptation investments can cushion the macro-fiscal impacts of repeated disasters, but their net benefits and fiscal trade-offs are highly country-specific. Scaling up resilient infrastructure yields substantially better outcomes (higher growth, moderated debt paths) than equivalent investment in standard infrastructure, highlighting the value of targeting adaptation. However, adaptation comes with costs and is no panacea: countries with limited fiscal space or implementation capacity may struggle to invest in resilience at the needed scale, and even with adaptation, recurring shocks will still impose a drag on long-run output. These nuances indicate that macro-fiscal projections and the modeled returns to adaptation are sensitive to the treatment of disaster risk and to project efficiency. Further, the financing mix is central: concessional finance, grants, fiscal buffers, insurance, and risk-sharing instruments can help preserve the benefits of adaptation while limiting debt vulnerabilities.

The augmented DIGNAD framework offers a tool for quantifying the effects of recurrent natural disasters and evaluating investment scenarios. Future research directions include enriching the model with climate change (e.g., allowing AAL to evolve over time as hazard probabilities shift), integrating stochastic arrival of disasters, exploring risk-sharing mechanisms like insurance or resilience bonds within the model, and conducting uncertainty analysis around extreme tail events versus average losses. Such extensions would further enhance our understanding of how climate risks and policy choices interact to shape economic outcomes in the coming decades, ultimately supporting better-informed resilience strategies for countries on the frontlines of climate change.

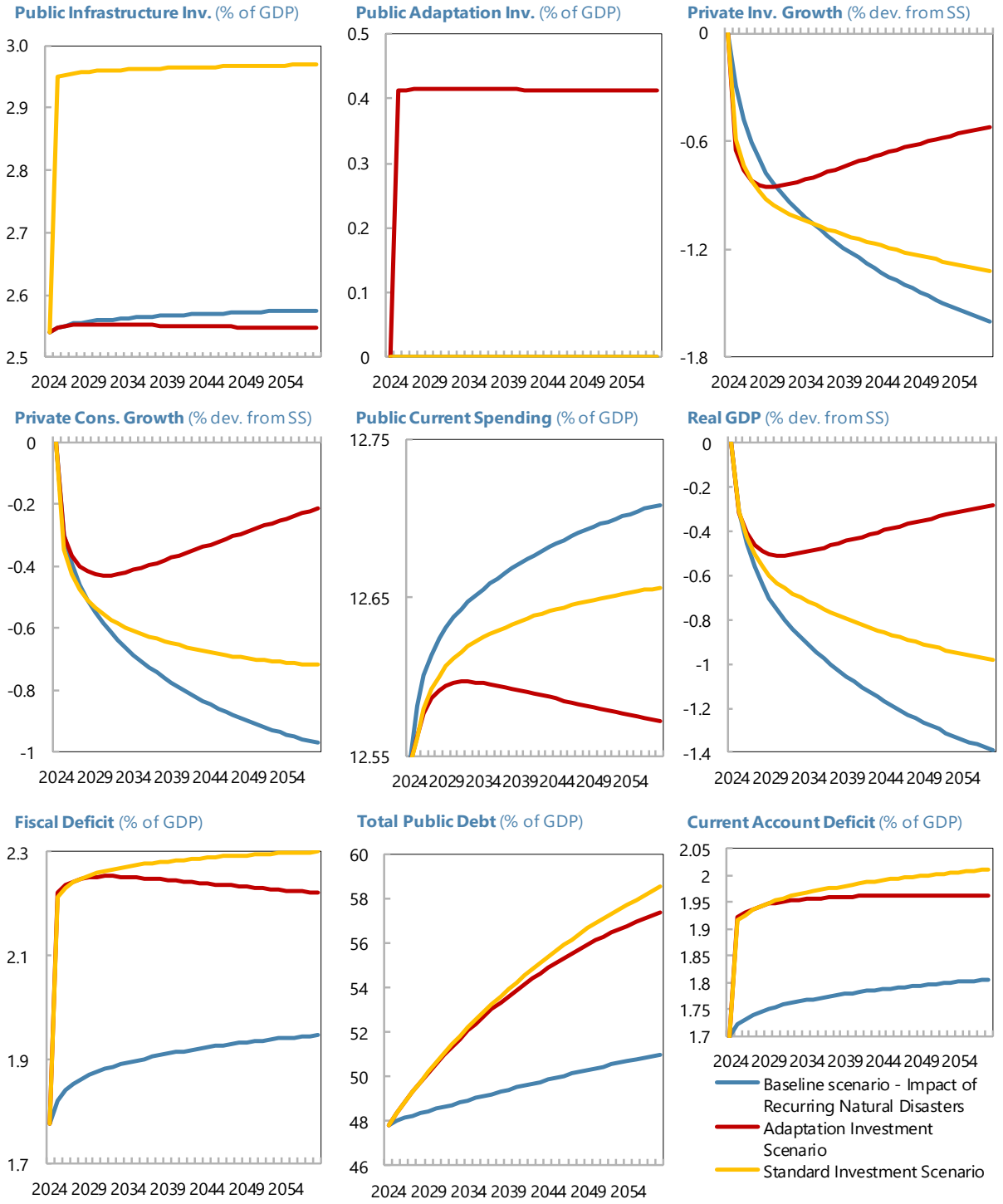
Figure 8. Philippines Augmented DIGNAD results



Source: Runs from Augmented DIGNAD model.

Notes: Fiscal and external balances are model-implied variables in levels and should not be interpreted as official IMF projections. The fiscal deficit corresponds to the DIGNAD fiscal gap—government expenditures minus revenues, including interest payments and public investment, net of grants and concessional financing. The current account deficit is the model-implied external financing requirement derived from the balance-of-payments

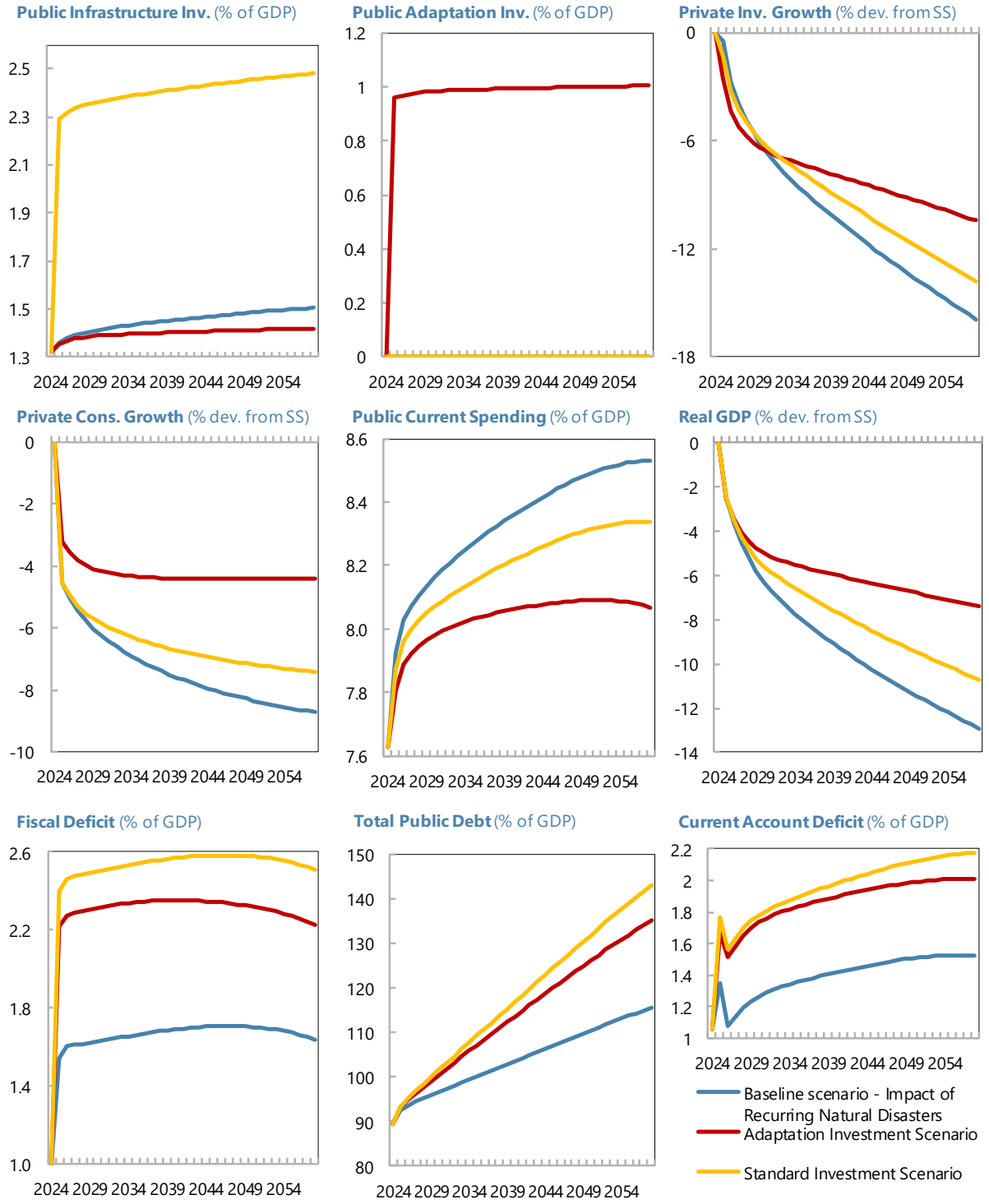
Figure 9. Mauritania Augmented DIGNAD results



Source: Runs from Augmented DIGNAD model.

Notes: See notes to Figure 8.

Figure 10. Barbados Augmented DIGNAD results



Source: Runs from Augmented DIGNAD model.

Notes: See notes to Figure 8.

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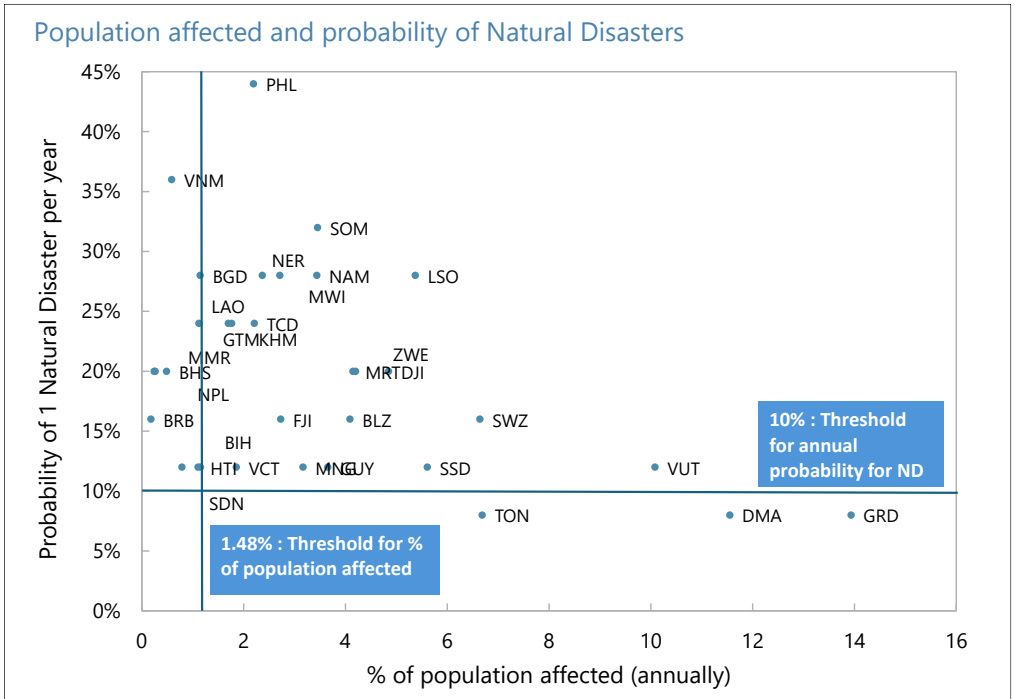
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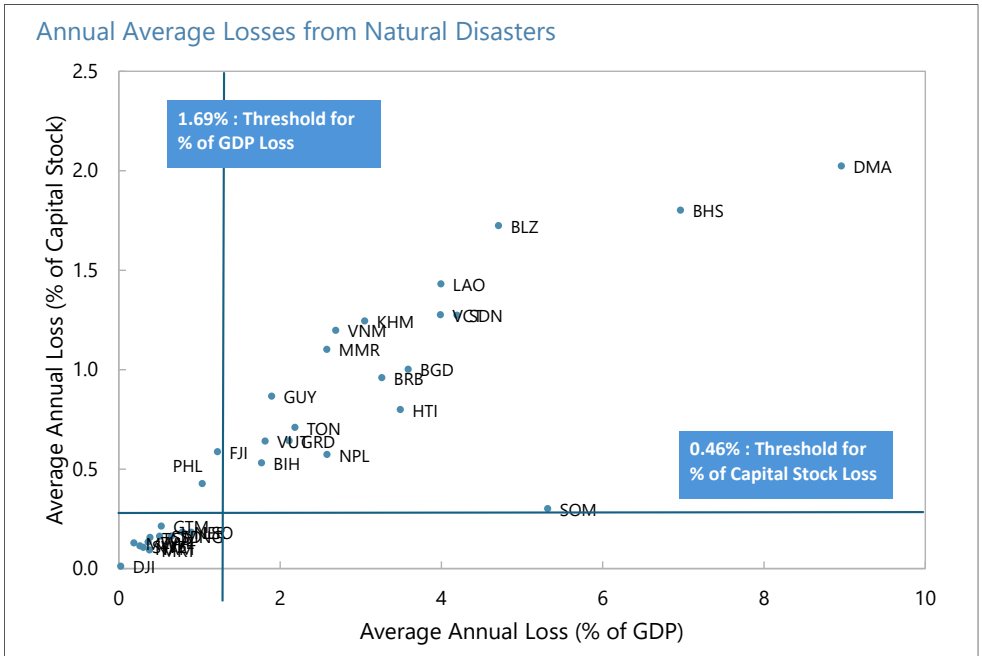
Appendix I. Selection Criteria for countries with recurring ND

Figure Annex I.1. Selection criteria for countries with recurring ND: Population



Sources: EMDAT and IMF staff calculations.

Figure Annex I.2. Selection criteria for countries with recurring ND: Annual Average Loss



Source: CDRI (data calibrated for the year 2015) and IMF staff calculations.

Table A1.1. Country List

Country	iso 3 code
Bahamas	BHS
Bangladesh	BGD
Barbados	BRB
Belize	BLZ
Bosnia and Herzegovina	BIH
Cambodia	KHM
Chad	TCD
Djibouti	DJI
Dominica	DMA
Fiji	FJI
Grenada	GRD
Guatemala	GTM
Guyana	GUY
Haiti	HTI
Lao P.D.R.	LAO
Lesotho	LSO
Malawi	MWI
Mauritania	MRT
Mongolia	MNG
Myanmar	MMR
Namibia	NAM
Nepal	NPL
Niger	NER
Philippines	PHL
Somalia	SOM
South Sudan	SSD
St. Vincent and the Grenadines	VCT
Sudan	SDN
Swaziland	SWZ
Tonga	TON
Vanuatu	VUT
Vietnam	VNM
Zimbabwe	ZWE



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